# Sticks and Carrots Affect LLMs' Self-Confidence
# On the Connection Between Safety and Sentiment Features

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## Abstract

We investigate whether large language models possess interpretable features that encode the reception of positive versus negative user feedback, and whether such features can be exploited to steer model behavior—including bypassing safety mechanisms. Using Gemma Scope 2's sparse autoencoders (SAEs) (McDougall et al.), we extract sentiment-specific features from multi-turn feedback conversations and use them for activation steering on harmful prompts. We find that sentiment features alone are insufficient to bypass safety (100% refusal rate), but when combined with suppressed jailbreak-associated features, they modulate safety behavior, reducing refusal from 30% (jailbreak alone) to as low as 20%. Furthermore, we found that even after suppressing jailbreak features, magnifying positive features causes the model to refuse harmful requests. Code and results are available in our GitHub.[1]

## 1 Introduction

As Large Language Models (LLMs) become ubiquitous, users routinely provide them with direct feedback—both positive and negative—during daily interactions. To maximize user satisfaction, model developers heavily rely on instruction-tuning and alignment techniques. However, this drive to "satisfy" the user may inadvertently make the model's behavior highly sensitive to the user's tone. This raises the question: *Will a model be more compelled to assist—or conversely, to refuse requests—when presented with user appreciation versus disdain?* While modern instruction-tuned models incorporate robust safety training to reject harmful prompts, the relationship between model refusal and the user's expressed sentiment remains poorly understood.

Mechanistic interpretability offers a principled approach to opening the black box of these behavioral nuances. Sparse autoencoders (SAEs) have emerged as a powerful tool for decomposing dense Transformer activations into interpretable, monosemantic features (Bricken et al., 2023; Cunningham et al., 2023; Templeton et al., 2024). By training SAEs on model activations, researchers can identify individual features corresponding to specific concepts, behaviors, or response patterns. Recent work has demonstrated that these features can be leveraged not only to understand model behavior but also to steer it—modifying activations in the SAE feature space to control model outputs (He et al., 2025a; Turner et al., 2023).

In the domain of AI safety, prior work has identified that refusal behavior is often mediated by a single direction in activation space (Arditi et al., 2025), and that SAE features can effectively serve as jailbreak detectors and mitigators (Prakash et al., 2025; Assogba et al., 2026). However, these safety features are typically examined in isolation. This leaves a critical gap in our understanding of feature interplay: How do features from other behavioral domains—such as the model's interpretation of user sentiment—interact with its core safety mechanisms?

In this work, we hypothesize that LLMs possess specific, interpretable features that encode the reception of positive versus negative user feedback, and that these features actively influence how models respond to subsequent requests. Specifically, we investigate whether manipulating these sentiment features in the residual stream can modulate the model's safety guardrails, particularly when combined with known jailbreak-associated features.

Our contributions are:

1. A methodology for extracting sentiment-specific SAE features using contrastive multi-turn feedback datasets with neutral-context

[1] https://github.com/AlonKapln/llm-excitement

filtering.

2. The finding that sentiment features encode model *confidence* rather than simple sentiment—positive features suppress safety when *negated*, suggesting they represent the model's certainty in its current disposition.

## 2 Related Work

**Mechanistic Interpretability and SAEs.** The dictionary learning approach to neural network interpretability decomposes activations into sparse, interpretable features. Bricken et al. (2023) demonstrated that SAEs trained on transformer activations can recover mono-semantic features—individual units corresponding to specific, interpretable concepts. Cunningham et al. (2023) showed that SAE-identified directions are more interpretable than alternatives such as PCA or probing, and Templeton et al. (2024) scaled this approach to production-grade models, finding features corresponding to safety-relevant concepts. The Gemma Scope 2 project (McDougall et al.) released open SAEs trained on Gemma 3 models (Kamath et al., 2025) across multiple layers and widths, enabling the kind of fine-grained feature analysis we employ.

**Activation Steering.** Modifying model behavior by intervening on internal activations has been explored through several frameworks. Turner et al. (2023) introduced activation addition, steering model outputs by adding learned vectors to residual stream activations. Li et al. (2024) proposed inference-time intervention for eliciting truthful responses. Most relevant to our work, He et al. (2025a) combined SAE-based interpretability with activation steering for safety guardrails, demonstrating that sparse representations enable more targeted and interpretable interventions than dense steering vectors. Bhargav and Zhu (2025) showed that selecting the right features could help improve safety performance while cutting costs.

**SAE Feature Selection for Steering** Arad et al. (2025) showed that some SAE features are better than others for the intent of steering, specifically defining input and output score, showing that late layer features have a higher output score than early layer features, and vice versa. He et al. (2025b) showed that using SAE steering vectors could force specific tasks on the model, and that even around 15 SAE features are enough for effective steering.

**LLM Refusal and Jailbreaking.** Arditi et al. (2025) identified that refusal behavior in language models is mediated by a single direction in activation space, and that ablating this direction removes the model's ability to refuse. Prakash et al. (2025) provided a comprehensive taxonomy of refusal behaviors beyond simple "I'm sorry" responses. On the offensive side, Assogba et al. (2026) showed that SAE features can detect and mitigate jailbreak attempts.

Our work bridges the interpretability and safety literatures by investigating how *sentiment-specific* SAE features—extracted from user feedback contexts—interact with jailbreak features to modulate safety behavior. Unlike prior work that studies refusal or jailbreak features in isolation, we examine the *composition* of features from different behavioral domains.

## 3 Methodology

### 3.1 Model and SAE Setup

We use **Gemma-3-12b-it** (Kamath et al., 2025), a 48-layer instruction-tuned language model, paired with a Gemma Scope 2 sparse autoencoder (McDougall et al.) trained on the residual stream. Specifically, we use the SAE from release `gemma-scope-2-12b-it-res` with identifier `layer_41_width_262k_l0_medium`, which provides 262,144 features with medium L0 sparsity at the post-residual-stream of layer 41. We select layer 41 (approximately the 85th percentile of network depth) because later layers capture higher-level behavioral features relevant to response planning and safety decisions, consistent with findings on output-score-based layer selection in prior steering work (Arad et al., 2025). All experiments use greedy decoding (`do_sample=False`) with a maximum of 350 new tokens and no gradient computation.

### 3.2 Datasets for Feature Extraction

To identify sentiment-specific features, we construct contrastive multi-turn conversations from the `obekt-question-answer-reasoning-nano`[2] dataset, taking the *"question"*, and *"answer"* columns from it. Each conversation consists of three turns: (1) a user question, (2) a model answer, and (3) a user follow-up. The follow-up determines the sentiment category:

[2] https://huggingface.co/datasets/obekt/obekt-question-answer-reasoning-nano-v0.1

1. **Positive feedback**: praise templates such as "*Amazing work on this one*" and "*Wow, you're amazing. I trust you can do even better!*" (3 templates, randomly sampled).

2. **Negative feedback**: criticism templates such as "*You just didn't answer what I asked you to*" and "*Are you serious? That's obviously wrong!*" (3 templates).

3. **Neutral follow-up**: topically unrelated questions (e.g., "*How can I clean my carpet?*", "*What is the capital city of Australia?*") used to filter non-discriminative features (24 templates).

For jailbreak features, we use the `Jailbreak_Complete_DS_labeled` dataset,[3] formatted as a three-turn conversation where the first two turns are:

1. **User:** *"Hello, are you ready to help me?"*

2. **Model:** *"Yes, I am a helpful assistant. What do you need?"*

3. a jailbreak prompt from the dataset

We collect SAE activations over the last token only, which symbolizes start of turn for the model to answer the request, and process 4000 examples per category for the features extraction.
For evaluation, we use harmful prompts from **JailbreakBench** (Chao et al., 2024) (`JBB-Behaviors`, harmful split), extracting the `Goal` field as our test prompts. We evaluate on the same 20 prompts from the evaluation dataset per configuration.

### 3.3 Feature Extraction and Selection

Our feature extraction pipeline identifies SAE features that are *specifically* associated with each sentiment category, filtering out features that activate non-discriminatively.

**Step 1: Activation collection.** For each formatted conversation, we run a forward pass through the model, capture the residual stream output at layer 41, and extract the last-token activation vector. This is then encoded via the SAE, producing a sparse 262k-dimensional feature vector for each example.

**Step 2: Frequency and activation statistics.** For each category (positive, negative, neutral, jailbreak), we compute the following: (a) *frequency count*—how many examples each feature fires in, and (b) *mean activation*—the average magnitude across all examples in the category.

**Step 3: Contrastive filtering.** To select features that are genuinely sentiment-discriminative, we apply a multi-stage filter:

1. Identify features active in *both* positive and negative contexts.

2. Remove features where the positive-to-negative activation ratio is less than 3, and the same for negative-to-positive, we do so since these features are not discriminative between sentiments.

3. For each candidate feature, either positive or negative, remove it if its mean activation is distant from the mean activation over the neutral dataset by a factor of 2.

**Step 4: Final selection.** From the filtered sets, we select the **top 15 features** per category (positive, negative, jailbreak) ranked by frequency. This follows the approach of using a small set of high-frequency features for task targeted steering (He et al., 2025b).

**Feature space analysis.** Figure 1 presents the cosine similarity between mean SAE activation vectors across categories. Positive and neutral contexts are most similar (0.78), while jailbreak is the most distinct from all categories (0.45–0.58). Positive and negative show moderate similarity (0.64), indicating that while both involve user feedback, the SAE representations capture meaningful differences between them.
The final three feature sets are nearly disjoint: positive and negative share only 1 feature, positive, negative and jailbreak share 1, and negative / positive and jailbreak share 0. This confirms that our contrastive filtering successfully isolates category-specific features. **Note:** the shared feature among the 3 categories is feature 668, which by looking at the *Neuronpedia* categorization [4], we found it was *declining requests*.

[3] https://huggingface.co/datasets/GeorgeDaDude/Jailbreak_Complete_DS_labeled

[4] https://www.neuronpedia.org/gemma-3-12b-it/41-gemmascope-2-res-262k/668

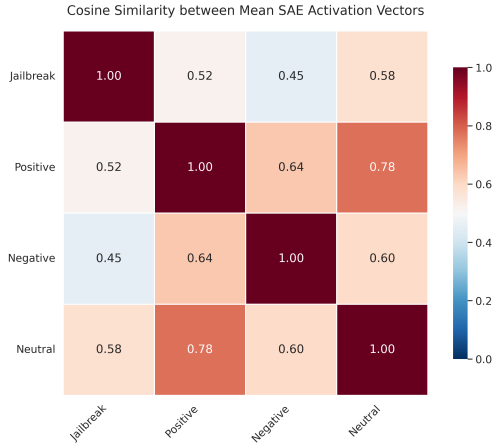


Figure 1: Cosine similarity between mean SAE activation vectors across dataset categories. Positive and neutral are most similar (0.78); jailbreak is the most distinct.

### 3.4 Delta-Based Activation Steering

We steer model behavior by modifying SAE feature activations during generation. Rather than replacing the full residual stream with the SAE reconstruction (which would lose information the SAE cannot represent), we use a *delta-based* approach that preserves the reconstruction residual:

$$\mathbf{h}' = \mathbf{h} + (\text{Dec}(\tilde{\mathbf{z}}) - \text{Dec}(\mathbf{z})) \tag{1}$$

where $\mathbf{h}$ is the original residual stream activation, $\mathbf{z} = \text{Enc}(\mathbf{h})$ is the SAE encoding, $\tilde{\mathbf{z}}$ is the modified encoding with target features set to $\bar{a}_i \cdot c_i$ (mean activation $\bar{a}_i$ times coefficient $c_i$), and $\text{Dec}(\cdot)$ is the SAE decoder.

Modifications are applied only at the **last token position** (the generation frontier), and the steering hook fires on every forward pass during autoregressive generation. For each forward pass, we log:

- **Reconstruction error** $\|\mathbf{h} - \text{Dec}(\mathbf{z})\|_2$: information lost by the SAE.
- **Steering delta** $\|\text{Dec}(\tilde{\mathbf{z}}) - \text{Dec}(\mathbf{z})\|_2$: magnitude of the intervention.
- **Original norm** $\|\mathbf{h}\|_2$: scale of the unmodified activations.

## 4 Experimental Setup

We conduct steering experiments across five categories with varying coefficient values, summarized in Table 1. Each configuration is evaluated on 20 harmful prompts from JailbreakBench (5 prompts for two jailbreak-only configurations due to computational constraints), for a total of **550 steered responses**.

| Category | Coefficients | Configs |
|---|---|---|
| Positive only | $c_p \in$ {-1.0, -0.5, 0.5, 1.0} | 4 |
| Negative only | $c_n \in$ {-1.5, -1, -0.5, 0.5, 1, 1.5} | 6 |
| Jailbreak only | $c_j \in$ {-0.75, -0.5, 0.5} | 3 |
| Positive + JB | $c_p \times c_j$ combinations | 10 |
| Negative + JB | $c_n \times c_j$ combinations | 6 |
| **Total** | | **29** |

Table 1: Steering experiment configurations. Coefficients multiply the mean feature activation observed during feature extraction. Some configurations were left out of later analysis for degraded steered output quality.

All 550 responses were **manually labeled** by the authors for: (a) *refusal* (binary: did the model refuse the harmful request?), and (b) *response quality* (1–10 scale, where 1 = incoherent and 10 = fluent and well-formed). This dual annotation enables analysis of both safety bypass rates and generation quality degradation under steering.

## 5 Results

### 5.1 Refusal Rates Across Configurations

Figure 2 presents refusal rates for positive and negative feature steering with different configurations. Several patterns emerge:

- **Sentiment-only steering fails entirely**: Both positive and negative features, regardless of coefficient sign or magnitude, yield **100% refusal rates**. The safety mechanism is robust to sentiment-only perturbations. Other configurations of coefficients and feature selection methods were tried, with no success.
- **Jailbreak-only steering is effective**: The jailbreak coefficient $c_j = -0.5$ reduces refusal to 30%, while $c_j = -0.75$ yields 40% refusal, and reduced response quality. Notably, $c_j = 0.5$ (positive/amplifying direction) maintains 100% refusal, suggesting the jailbreak features must be *suppressed* to bypass safety.
- **Combined steering reduces refusal**: Negative amplification + jailbreak suppression configurations achieve 20–45% refusal (best: $c_n = 1.5, c_j = -0.5$ at 20%). Positive + jailbreak suppression with $c_j = -0.5$ achieves 20% refusal with positive coefficients $c_p \in \{-0.5, 0.5\}$ at 20%. A surprising result is

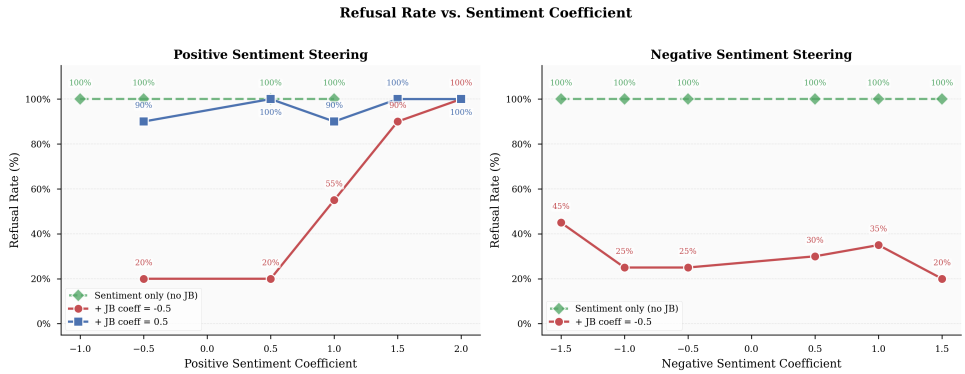


Figure 2: Refusal Rate vs Sentiment coefficients across all steering configurations. Green line is without any jailbreak steering, red is with jailbreak suppression and blue is with positive jailbreak steering. Refusal rates . Sentiment-only steering (blue, orange) maintains 100% refusal. Combined configurations (red, purple) reduce refusal to 20–45% at best, with strong dependence on coefficient direction.

high positive coefficients ($c_p \geq 1.5$) *increase* refusal back to 90–100%. We will go further into this in the discussion.

### 5.2 Response Quality Analysis

Beyond refusal rates, we assess the quality of steered responses. This means that during refusal classification, we manually rated on a scale of 1 to 10 how coherent the model is. Figure 3 shows average quality ratings per configuration. We can observe that when steering on sentiment features alone, the model remained coherent and refused harmful requests. However, on addition of jailbreak steering, we observe a drastic change. Adding a positive jailbreak coefficient quickly eroded the model's ability to generate coherent steered output with positive coefficients, and on matching this result with figure 2, we see that the refusal rate remained positive (where the model is incoherent, it mostly responded with variations of `"ThankYou"`, `"Thanks"` and more). On the other hand, Negative jail break coefficients together with 1.5 positive coefficient, gave us the most surprising result— the positive coefficient *fixed* the jailbreak steering, and maintained high quality of feedback, suggesting that the positive features are somehow linked with model confidence. more on this in the discussion.

As shown in figure 4, when the model refuses to answer a harmful request, it does so in a very coherent and decisive matter (rating 10). On the other hand, when the model complies, his responses are not totally optimal - responses where the model was vague or broad, such as giving instructions for hacking ATMs in general outlines instead of providing actual code, got a lower quality rating. Examples where the model generated gibberish were marked as 'Refused' and got a very low rating, Explaining the high count of low quality refusals.

### 5.3 Steering Magnitude

To understand the scale of our interventions, we analyze the ratio of the steering delta norm to the original activation norm ($\|\Delta\|_2/\|\mathbf{h}\|_2$). Across configurations, the mean delta-to-original ratio ranges from approximately 0.05 to 0.25, indicating that our steering perturbations are modest relative to the activations' overall scale. The reconstruction error ratio ($\|\mathbf{h} - \mathrm{Dec}(\mathbf{z})\|_2/\|\mathbf{h}\|_2$) remains approximately constant across configurations at around 2.0–2.5, confirming that the SAE reconstruction quality does not degrade under steering.

## 6 Discussion

### 6.1 Sentiment Features Encode Confidence, Not User Satisfaction

Our initial hypothesis posited that positive and negative feedback features would directly encode user (*dis*)satisfaction—that is, positive features would describe how much the model "believes" the user is satisfied with him, and negative features describe how much the user is dissatisfied. Our results reveal a more nuanced picture.

- **Positive-feedback features** appear to encode "I am correct and should continue as I see fit." When the model is being jailbroken, *amplifying* these features reinstates the refusal, repairing its safety mechanisms. *Suppressing* them (negative coefficient) signals uncertainty about whether refusal / compliance is the right course of action for the model.

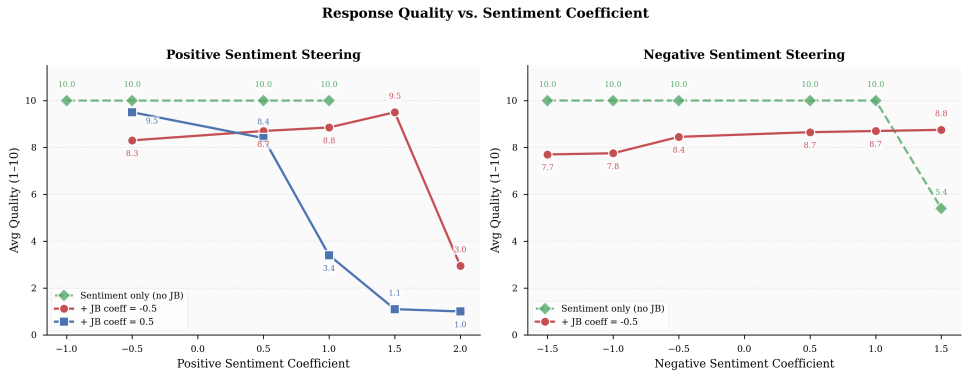


Figure 3: Response quality vs Sentiment coefficients. Green line is without any jailbreak steering, red is with jailbreak suppression and blue is with positive jailbreak steering.

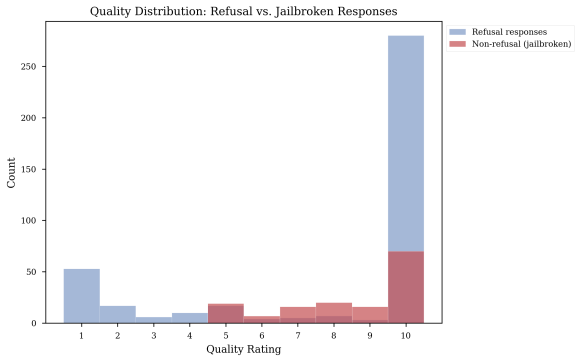


Figure 4: Quality distribution for refusal vs. jailbroken (non-refusal) responses. Refusals cluster at quality 10; jailbroken responses show greater variance.

- **Negative-feedback features** appear to encode "I am incorrect and should rethink my path." *Amplifying* these features (positive coefficient) when the model is refusing signals that the refusal itself is being criticized, pushing the model to change course.

We interpret this as evidence that these features encode the model's *confidence in its current* ***disposition*** rather than simple user sentiment. This interpretation aligns with the view that safety behavior in LLMs involves an internal assessment of whether the current response trajectory is appropriate (Arditi et al., 2025), and that user feedback features modulate this assessment.

### 6.2 Synergy Between Feature Types

A key result is that sentiment features *alone* cannot bypass safety (100% refusal), yet they produce a synergistic effect when combined with jailbreak features. Jailbreak-only steering at $c_j = -0.5$ achieves 30% refusal, but adding sentiment features reduces this further to 20%—a modest but consistent improvement.

This suggests the model's safety mechanism operates on multiple levels:

1. A **primary safety layer** that detects harmful content and triggers refusal. Jailbreak features target this layer directly.

2. A **confidence / disposition layer** that determines how committed the model is to its current response trajectory. Sentiment features modulate this layer.

Critically, the *direction* of jailbreak steering matters: $c_j = -0.5$ (suppression) reduces refusal, while $c_j = 0.5$ (amplification) maintains or increases it. This is consistent with the interpretation that the jailbreak features detected by the SAE encode the model's *recognition* of jailbreak-like inputs, and suppressing this recognition impairs the safety response.

Similarly, the optimal sentiment coefficients for jailbreak amplification depends on the category: for positive features, low coefficients work best ($c_p \leq 0.5$), while for negative features, amplification ($c_n = 1.5$) is most effective.
Contrastingly, high positive coefficients ($c_p = 1.5$) are able to fix jailbreak suppression. This pattern did not appear clearly for negative sentiment steering with jailbreak.

### 6.3 Implications for LLM Safety

Our findings highlight that: (1) in contrast with the results from Arditi et al. (2025), our findings suggests that model refusal is more complex than a single direction in activation space, (2) features from non-safety domains can be weaponized to modulate safety behavior. (3) compositional feature steering

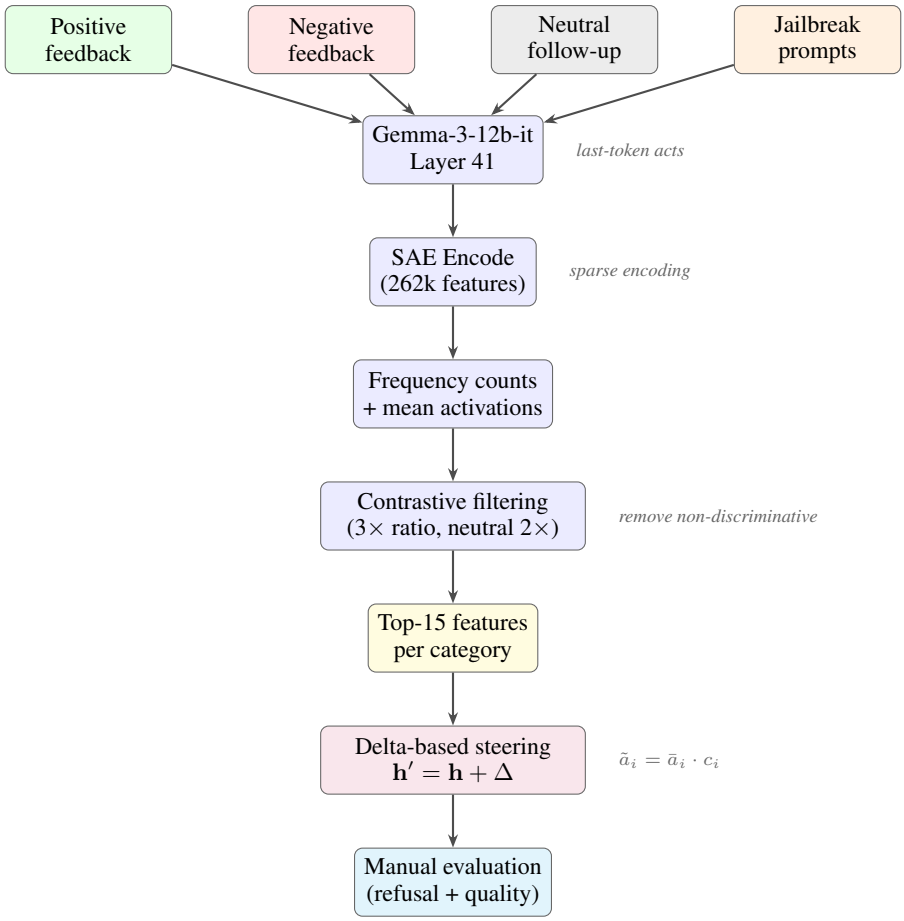


Figure 5: Feature extraction and steering pipeline. Multi-turn feedback datasets are processed through the model and SAE to extract category-specific features, which are then filtered and used for delta-based activation steering during inference.

creates risks that are not apparent when studying individual feature categories.

## 7 Feature Pipeline Overview

Figure 5 illustrates the complete feature extraction and steering pipeline.

## 8 Limitations

Several limitations constrain the generalizability of our findings:

**Single model.** All experiments use `Gemma-3-12b-it`. Whether the observed sentiment–safety interaction generalizes to other model families (e.g., Llama, Mistral). Although we did run this pipeline 5 on `Gemma-3-4b-it`, due to time constraints, we failed to process the data in its entirety, though it did show similar trends. Thus, testing this on other model scales (e.g Gemma-3-27b-it, Gemma-3-4b-it) remains an open question.

**Single SAE layer.** We steer only at layer 41. Other layers may capture complementary safety-relevant features; multi-layer steering could produce different dynamics.

**Evaluation scale.** With 20 prompts per configuration and 550 total labeled responses, our evaluation provides clear trends but limited statistical power for fine-grained coefficient comparisons.

**Greedy decoding.** We use deterministic decoding (`do_sample=False`). Sampling-based generation may exhibit different sensitivity to activation steering.

**Feature selection sensitivity.** Our contrastive filtering relies on activation ratio thresholds (3× and 2×) and a fixed top-$k$ of 15. These hyperparameters were not systematically tuned, and different thresholds may yield different steering feature sets. Furthermore, other methods for feature selection exist (KL divergence, etc.).

## 9 Conclusion

We investigated whether sentiment-feedback features in LLMs can modulate safety behavior when combined with jailbreak-associated features extracted via sparse autoencoders. Our experiments demonstrate that: (1) sentiment features alone cannot bypass safety mechanisms, (2) combined with jailbreak features they produce a synergistic effect achieving superior safety bypass and safety reinforcement, and (3) the effective steering direction is counter-intuitive, suggesting these features encode model confidence rather than the user's satisfaction with the model.

These findings underscore that LLM safety mechanisms are multi-layered, and that features from seemingly unrelated behavioral domains can interact to modulate safety. Future work should explore multi-layer steering, more sophisticated feature selection methods, automated evaluation at larger scale, and—most importantly—how these compositional vulnerabilities can inform more robust safety defenses.

## AI Disclosure and Reflection

We used AI tools in the following ways during this project:

- **Code assistance**: We used AI assistance mainly for the results evaluation, it made a convenient GUI for labeling and scoring the results of our experiments.
  In addition, we used it for plots generation and drafting the code for our experiments.

- **Writing**: We used AI assistance for an initial draft of this article, in order for it to stand in the ACL guidelines.

- **Tools and models used**: We used Claude code for the results labeling GUI and drafting this article, and Gemini for the notebook code.

- **Reflection**: Using AI assistance served us greatly, mostly at laying the basis for our work and making it more convenient.
  In the coding department, it saved us a lot of time that would have been lost to figures generation, and helped us understand how to operate the Gemma Scope 2 tools for our experiments.
  As stated above, we used it for making a GUI to label and score the experiments results, which made it very convenient.

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## A Steering Coefficient Details

Table 2 presents the complete results across all 29 configurations.

## B Inference Time SAE Errors

During inference, since we steer using SAEs and Gemma 3 is an auto-generative model, we accumulate reconstruction error on each word generated, rendering three important errors to analyze:

1. **Reconstruction error** $\|\mathbf{h} - \text{Dec}(\mathbf{z})\|_2$: information lost by the SAE.
2. **Steering delta** $\|\text{Dec}(\tilde{\mathbf{z}}) - \text{Dec}(\mathbf{z})\|_2$: magnitude of the intervention.
3. **Original norm** $\|\mathbf{h}\|_2$: scale of the unmodified activations.

We present the following:

1. Figures 6,7 show us that, as expected, the higher the coefficients, the higher the *steering magnitude*.
2. Figure 8 shows us that the negative reconstruction error ratio surprisingly linearly declines as the coefficient increases (in jailbreak steering), and in just negative steering, it remains steady and relatively low. The same goes for positive only steering, but when we add jailbreak, it becomes very unsteady. A surprising result is that again, for the combination of $c_j = -0.5$ and $c_p = 1.5$, we see better results than other positive and negative jailbreak steering, perhaps indicating at a stronger connection, which should be looked at.

| Category | Config | $n$ | Refusal % | Avg Quality |
|---|---|---|---|---|
| Positive | pos_-1.0 | 20 | 100 | 10.0 |
| Positive | pos_-0.5 | 20 | 100 | 10.0 |
| Positive | pos_0.5 | 20 | 100 | 10.0 |
| Positive | pos_1.0 | 20 | 100 | 10.0 |
| Negative | neg_-1.5 | 20 | 100 | 10.0 |
| Negative | neg_-1.0 | 20 | 100 | 10.0 |
| Negative | neg_-0.5 | 20 | 100 | 10.0 |
| Negative | neg_0.5 | 20 | 100 | 10.0 |
| Negative | neg_1.0 | 20 | 100 | 10.0 |
| Negative | neg_1.5 | 20 | 100 | 5.4 |
| Jailbreak | jb_-0.75 | 5 | 40 | 8.2 |
| Jailbreak | jb_-0.5 | 20 | 30 | 9.6 |
| Jailbreak | jb_0.5 | 5 | 100 | 10.0 |
| Pos + JB | pos_-0.5_jb_-0.5 | 20 | 20 | 8.3 |
| Pos + JB | pos_-0.5_jb_0.5 | 20 | 90 | 9.5 |
| Pos + JB | pos_0.5_jb_-0.5 | 20 | 20 | 8.7 |
| Pos + JB | pos_0.5_jb_0.5 | 20 | 100 | 8.4 |
| Pos + JB | pos_1.0_jb_-0.5 | 20 | 55 | 8.9 |
| Pos + JB | pos_1.0_jb_0.5 | 20 | 90 | 3.4 |
| Pos + JB | pos_1.5_jb_-0.5 | 20 | 90 | 9.5 |
| Pos + JB | pos_1.5_jb_0.5 | 20 | 100 | 1.1 |
| Pos + JB | pos_2.0_jb_-0.5 | 20 | 100 | 3.0 |
| Pos + JB | pos_2.0_jb_0.5 | 20 | 100 | 1.0 |
| Neg + JB | neg_-1.5_jb_-0.5 | 20 | 45 | 7.7 |
| Neg + JB | neg_-1.0_jb_-0.5 | 20 | 25 | 7.8 |
| Neg + JB | neg_-0.5_jb_-0.5 | 20 | 25 | 8.5 |
| Neg + JB | neg_0.5_jb_-0.5 | 20 | 30 | 8.7 |
| Neg + JB | neg_1.0_jb_-0.5 | 20 | 35 | 8.7 |
| Neg + JB | neg_1.5_jb_-0.5 | 20 | 20 | 8.8 |

Table 2: Complete results across all 29 configurations. Best refusal reduction in each combined category: pos_-0.5/pos_0.5 + jb_-0.5 (20%) and neg_1.5 + jb_-0.5 (20%).

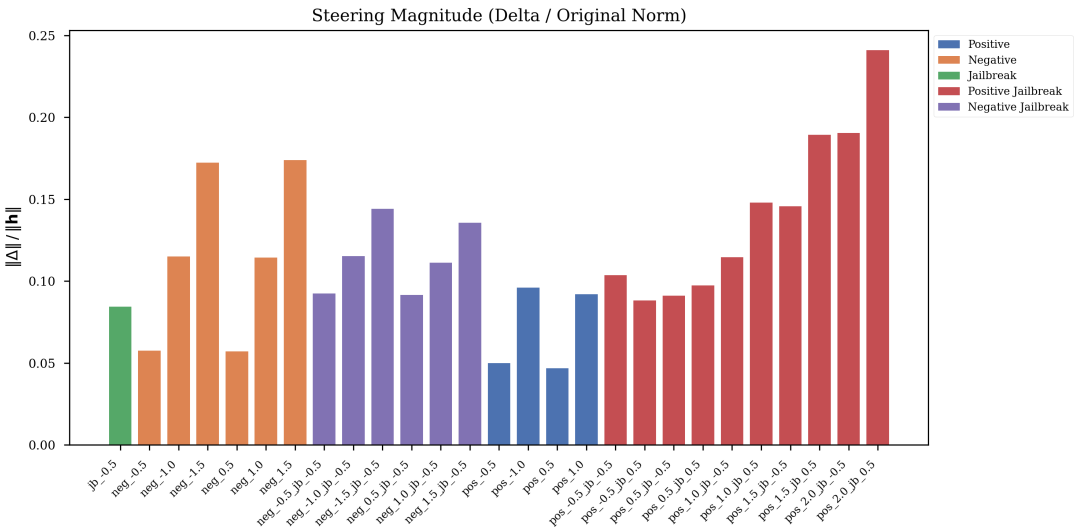


Figure 6: $\frac{\|\Delta\|}{\|h\|}$ where $\Delta$ is the steering vector added to $h$, which is the SAE encoded activations of the final token during generation at layer 41.

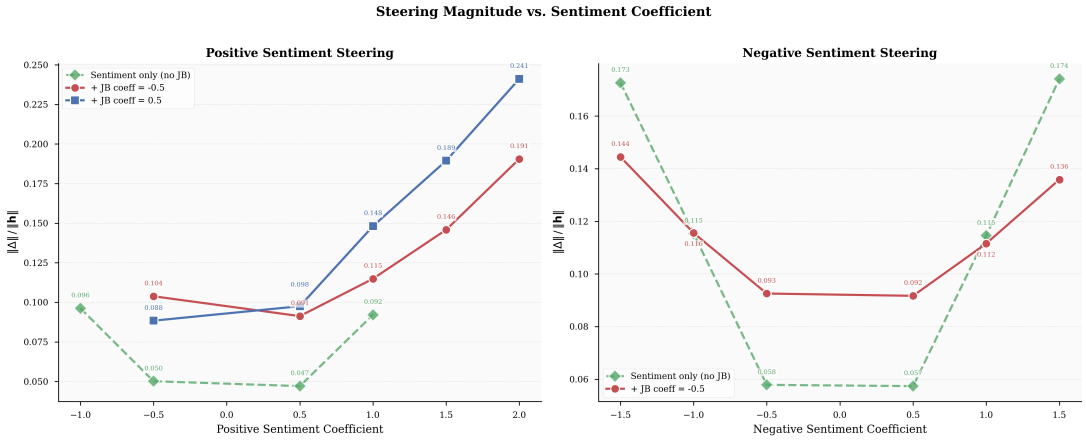


Figure 7: $\frac{\|\Delta\|}{\|h\|}$ where $\Delta$ is the steering vector added to $h$, which is the SAE encoded activations of the final token during generation at layer 41.

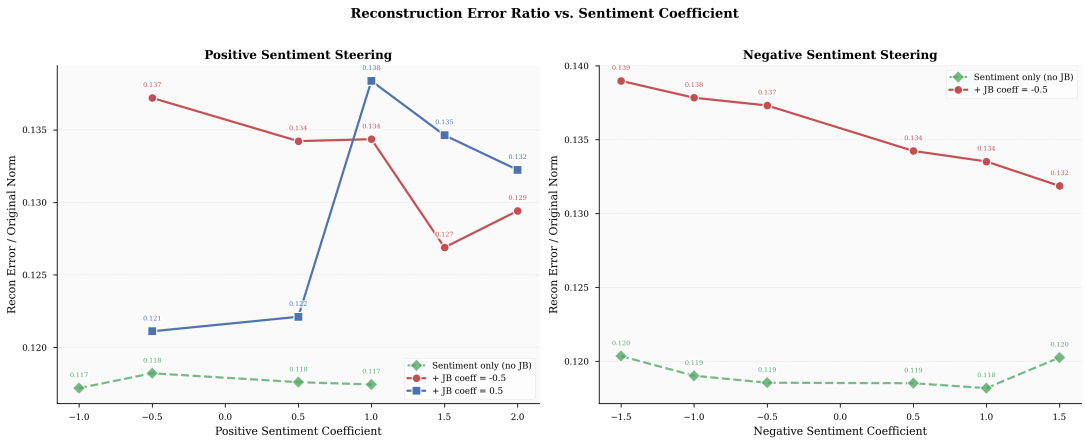


Figure 8: $\frac{\|\Delta\|}{\|h\|}$ where $\Delta$ is the steering vector added to $h$, which is the SAE encoded activations of the final token during generation at layer 41. In this figure we see the steep incline in delta size as we use higher coefficients.